



WAZUH SIEM INSTALLATION AND CONFIGURATION STEPS

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Wazuh SIEM Installation and Configuration Guide

(VirtualBox Network Setup to Agent Integration)

1. Objective

The objective of this document is to provide a complete step-by-step procedure for setting up a Wazuh SIEM environment using VirtualBox. This includes configuring the network, installing the Wazuh server, registering and configuring a Windows agent, and verifying log collection through the Wazuh dashboard.

2. Lab Architecture

Machine	Operating System	Role	IP Address
Wazuh Server	Ubuntu 22.04	Wazuh Manager + Dashboard	192.168.56.10
Windows Server	Windows Server	Wazuh Agent	192.168.56.20

3. VirtualBox Setup

3.1 Install VirtualBox

- Download and install VirtualBox and Extension Pack from the official website.

3.2 Create Virtual Machines

Wazuh Server (Ubuntu)

- RAM: Minimum 4 GB (8 GB recommended)
- CPU: 2 cores
- Storage: 50 GB
- Network Adapter: Host-Only Adapter

Windows Server (Agent)

- RAM: Minimum 4 GB
- Storage: 40 GB
- Network Adapter: Host-Only Adapter

4. Network Configuration

4.1 Create Host-Only Network

- Open VirtualBox → Tools → Host Network Manager
- Create a network with the following configuration:
 - Network: 192.168.56.0/24
 - Gateway: 192.168.56.1

4.2 Assign Static IP Address

Ubuntu (Wazuh Server)

Edit netplan configuration:

```
sudo nano /etc/netplan/00-installer-config.yaml
```

Example configuration:

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.56.10/24
      gateway4: 192.168.56.1
      nameservers:
        addresses: [8.8.8.8]
```

Apply changes:

```
sudo netplan apply
```

Windows Server

Set static IP manually:

- IP Address: 192.168.56.20
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.56.1
- DNS: 8.8.8.8

4.3 Test Connectivity

From Windows:

```
ping 192.168.56.10
```

5. Install Wazuh Server

5.1 Update System

```
sudo apt update && sudo apt upgrade -y
```

5.2 Install Wazuh (All-in-One)

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh  
sudo bash wazuh-install.sh -a
```

5.3 Verify Services

```
sudo systemctl status wazuh-manager  
sudo systemctl status wazuh-dashboard
```

6. Configure Firewall on Wazuh Server

```
sudo ufw allow 1514/tcp  
sudo ufw allow 1515/tcp  
sudo ufw allow 55000/tcp  
sudo ufw allow 443/tcp  
sudo ufw enable
```

7. Access Wazuh Dashboard

Open browser:

<https://192.168.56.10> [here the ip used will be the Wazuh server ip]

Accept SSL warning and log in using credentials displayed during installation.

8. Agent Registration on Wazuh Server

Run:

```
sudo /var/ossec/bin/manage_agents
```

Steps:

1. Press **A** to add agent
 2. Enter:
 - Name: windows-agent
 - IP: 192.168.56.20
 3. Press **E** to extract key
 4. Copy the generated key
 5. Press **Q** to exit
-

9. Install Wazuh Agent on Windows Server

9.1 Download Agent

Download from the official Wazuh website.

9.2 Install Agent

During installation:

- Enter Wazuh Server IP:
 - 192.168.56.10
-

10. Configure Agent

10.1 Verify Manager IP

Edit:

```
C:\Program Files (x86)\ossec-agent\ossec.conf
```

Ensure:

```
<client>
  <server>
    <address>192.168.56.10</address>
  </server>
</client>
```

10.2 Import Agent Key

Open Command Prompt as Administrator:

```
"C:\Program Files (x86)\ossec-agent\manage_agents.exe"
```

- Press **I**
- Paste the key copied from the server

10.3 Restart Agent Service

```
net stop wazuh-agent  
net start wazuh-agent
```

11. Verify Agent Connection

On Wazuh Server:

```
sudo /var/ossec/bin/agent_control -l
```

Agent status should appear as **Active**

12. Enable Windows Log Collection

Edit:

```
C:\Program Files (x86)\ossec-agent\ossec.conf
```

Add:

```
<localfile>  
  <location>Security</location>  
  <log_format>eventchannel</log_format>  
</localfile>
```

Restart service:

```
net stop wazuh-agent  
net start wazuh-agent
```

13. Verify Logs in Dashboard

- Open Wazuh Dashboard
 - Navigate to Security Events
 - Confirm logs are being displayed
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14. Time Synchronization

On Ubuntu:

```
timedatectl set-ntp true
```

On Windows:

- Enable automatic time synchronization
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15. Troubleshooting

Check Wazuh Logs:

```
sudo tail -f /var/ossec/logs/ossec.log
```

Restart Services:

```
sudo systemctl restart wazuh-manager  
sudo systemctl restart wazuh-dashboard
```

16. Final Verification Checklist

- Virtual machines are properly configured
 - Network connectivity is established
 - Wazuh server is accessible via browser
 - Agent is registered and active
 - Logs are visible in dashboard
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17. Conclusion

The Wazuh SIEM environment has been successfully configured using VirtualBox with proper network setup, server installation, and agent integration. The system is now capable of centralized log monitoring and provides a foundation for further security analysis and threat detection.